

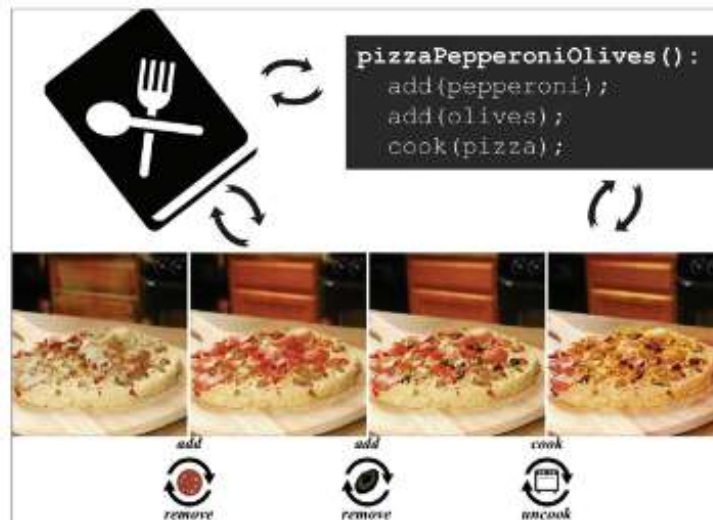


2016: Google's AI AlphaGo beat the professional Go player Lee Sedol convincingly. Go is considered a difficult game, so this victory by AI is considered particularly noteworthy and marks the progress in the machine learning abilities of AI.

2018: Google Duplex was announced by CEO Sundar Pichai. Duplex is a virtual assistant and can make reservations at restaurants and hairdressers on our behalf using a realistic-sounding artificial voice.



Have you heard about an AI that is a Pizza chef? MIT scientists have created a GAN AI-based model called PizzaGAN.



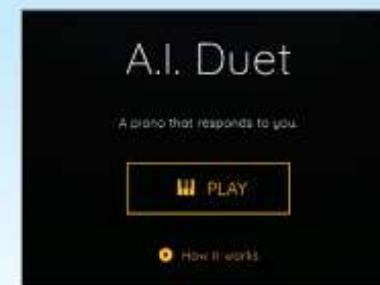
PizzaGAN is shown many images of the same pizza. It first identifies the toppings and the sequence of layers and then follows the sequence to make a pizza.



KiBo's Activity 2

Let us see if you can enjoy playing a duet with a GAN-based AI application built by Yotam Mann with friends at Google.

- ① Open a web browser on your computer and visit the following site:
<https://experiments.withgoogle.com/ai/ai-duet/view/>
- ② The A.I. Duet experiment screen will open, containing the **PLAY** button as shown.
- ③ Select the **PLAY** button and a screen with piano keys will appear.
- ④ Click on the keys you wish to play by using the mouse. The AI will respond by playing some keys of its own, thus collaborating with you in playing a duet. The blue colour represents the notes that you generate, whereas the yellow ones are generated by the AI application.



A.I. Duet Experiment Screen



A.I. Duet Screen Showing the Keys Played

So, did you enjoy playing the duet with the AI? That will give you the answer to **Query 2**.

Current Trends in AI

AI is developing at a rapid pace and at a remarkable level and is now being integrated into almost every sector. Companies like Google, Facebook, IBM, and Amazon are working with AI and creating amazing devices. Artificial intelligence is set to transform not only global industries and businesses but also the way we live. In 2021, the global pandemic resulted in the emergence of new forms of business issues. Companies have responded by developing new AI-powered products and solutions to address these issues. Thus, AI is set to increase at a phenomenal rate in the coming years.





Kibo's Query 1

Do you think the following sci-fi technologies will become a reality in the future? Which one of these would you like to experience in reality and why?

- ☐ Invisibility
- ☐ Teleportation (Instant transportation from one place to another)
- ☐ Time travel
- ☐ Space travel to other galaxies
- ☐ None of these

The First AI Winter (1974-1980)

The period between 1974 to 1980 was the first AI winter because there was reduced funding and interest in artificial intelligence research during this period.

A Boom of AI (1980-1987)

1980: AI came back into the forefront with a growing interest in **Expert System**. An expert system is a computer program that uses artificial intelligence to simulate the judgment and behaviour of a human who has expert knowledge and experience in a particular field. It performs specialised tasks. For example, an AI which is an expert in diagnosis of disease, similar to a doctor.



Kibo's Fun Fact

The first example of an expert system was the DENDRAL (acronym for the term 'Dendritic Algorithm'), developed in 1965 by **Edward Feigenbaum**, also referred to as the "father of expert systems". This program was developed to map the structures of molecules to help chemists identify unknown organic molecules.



The Second AI Winter (1987-1993)

Between the years 1987 and 1993, the world experienced the second AI Winter when investors and government stopped funding AI research as the expert systems like XCON proved too expensive to maintain. Governments reduced their investments because of the fear of the lack of returns and the focus shifted on desktop computers, which were more user-friendly.

The Era of Intelligent Devices and Services (1993-2011)

1997: **IBM Deep Blue**, an AI-based program, became the first computer to beat the world chess champion, **Gary Kasparov**. This win was an important milestone and was instrumental in launching the IBM supercomputer program that has paved the way for some of the leading-edge AI technology available in the world today.

2002: For the first time, AI entered the home in the form of **Roomba**, a vacuum cleaner launched by **iRobot**. It used artificial intelligence to scan the room size, identify obstacles, and remember the most efficient routes for cleaning.



2006: AI entered the business world in the year 2006. Companies like Facebook, Twitter, and Netflix have also started using AI.

2009: Google launched its self-driving car technology, with **Waymo**, its first self-driving car.



Waymo



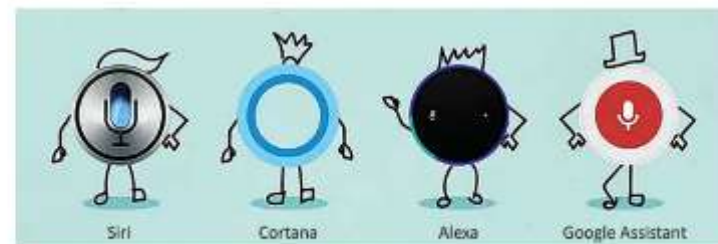
Find out the details of three games where AI has conquered human counterparts. One such link is given below:

<https://bdtechtalks.com/2018/07/02/ai-plays-chess-go-poker-video-games/>

Name of the Game	Name of the AI System	Year
1. _____	_____	_____
2. _____	_____	_____
3. _____	_____	_____

Exponential Growth and Development of Artificial Intelligence (2011-present)

2011-2014: In the year 2011, **Project Debater**, the first AI system that can debate with humans on complex topics, debated with two master debaters and performed very well. Its goal is to help people build convincing arguments and make well-informed decisions. During this period, personal digital assistants like Siri, Google Assistant, and Cortana were introduced. These assistants used speech recognition to answer questions and perform tasks.



Digital Assistants

2014: **Ian Goodfellow** invented the **Generative Adversarial Network (GAN)**, which is a set of two competing neural networks that are used to generate accurate output like images, videos, and voice. It is like two humans using their intelligence to compete with each other and come up with the best solution. Let us consider an example of how GAN works. If you train a GAN on a lot of dog photos, it can combine these to produce a convincing fake dog photo. GAN is currently being used in several art and graphic projects.





It's Story Time! Ancient Greek mythology, first narrated by Hesoid in 700 B.C., talks about a giant killer robot called **Talos**, forged by the Greek God **Hephaestus** to protect the island of Crete. This offers one of the earliest conceptions of a robot. To learn about the fascinating story of Talos and the Argonauts and these earlier conceptions of AI, follow the YouTube link:

<https://www.youtube.com/watch?v=vVTA-E3G8bQ>

Write in brief about how Talos was finally defeated by the Argonauts.

The Advent of the Age of Information Technology

The first mechanical calculating machine was built by French mathematician and inventor **Blaise Pascal** in 1642. **Charles Babbage** and **Ada Lovelace** created the first computer, a programmable machine, in 1837. After this breakthrough, the quest was on to make computing machines better at performing tasks similar to humans.

The Birth of Artificial Intelligence (1943-1952)

The study of the human brain is thousands of years old. With the advent of modern electronics, scientists were of the view that the thinking process of the brain could be simulated artificially. Modern AI began during the 1940s with the goal of solving complex mathematical problems and creating 'thinking machines.'

1943: The first step toward artificial neural networks came in 1943, when **Warren McCulloch**, a neurophysiologist, and a young mathematician, **Walter Pitts**, wrote a paper on how neurons work. They modelled a simple neural network, i.e., a model of an Artificial Neural Network (ANN) with electrical circuits based on the working of the human brain.

1950: **Alan Turing** was an English mathematician who pioneered machine learning in 1950. In his paper 'Computing Machinery and Intelligence', Alan Turing discussed how to build intelligent machines. The Turing test formulated by him can check the machine's ability to exhibit intelligent behaviour similar to human intelligence.



KiBo's Fun Fact

Even before the word 'Artificial Intelligence' was coined, one of the most popular artificially intelligent robots was shown as a character—the 'heartless' Tin man from the book '**Wizard of Oz**', which was then made into a very popular movie in 1939.



Discover about some well-known movies from the 20th century that featured characters with artificial intelligence.

Name of movie

Character displaying AI

1.

2.

The Maturation of Artificial Intelligence (1952-1956)

1955: The 'first artificial intelligence program' or 'thinking machine' called **Logic Theorist** was created by **Allen Newell** and **Herbert A. Simon**. It was the first program to perform automated reasoning and succeeded in showing that AI could prove mathematical theorems like any talented mathematician.

1956: The word 'Artificial Intelligence' was first coined by the American computer scientist **John McCarthy** at a conference at Dartmouth. The objective of the conference was to find ways to make a machine that could reason like a human and be capable of abstract thought, problem-solving, and self-improvement. This laid the foundation of AI as an academic field.

High-level computer languages such as FORTRAN, LISP, or COBOL were developed during this period. There was a very high enthusiasm for AI at this time.

The Golden Years of AI (1956-1974)

1966: **Joseph Weizenbaum** created the first chatbot, **ELIZA**, in 1966, which could trick its users into believing that they were having a conversation with a real human being. **ELIZA** was designed to imitate a psychotherapist who would ask questions and even respond like a human therapist. It succeeded in tricking at least 33% of the people into believing that it was human.

1972: The first intelligent humanoid robot, **WABOT-1** (WAseda roBOT), was built in Japan. A humanoid robot, also called an **android**, is a robot that resembles a human body in shape (anthropomorphic).



WABOT



KiBo's Fun Fact

The first chatbot, **ELIZA**, was created in 1966, but the term 'ChatterBot' existed in 1994, which naturally evolved into the word 'Chatbot'. It was coined by **Michael Mauldin**, creator of the first Verbot Julia, to describe these conversational programs.

